

# Session 2: How Software Works.

## Task 1: The Three Main Layers of Software Architecture

- **UI Layer (Presentation Layer):** The visual interface and screens that users see and interact with directly, such as buttons, text fields, and menus.
- **Business Logic Layer (Application Layer):** The "brains" of the software that processes data, enforces rules, and performs calculations behind the scenes.
- **Database Layer (Data Storage Layer):** The secure warehouse where all the application's permanent data and records are stored and organized.

## Task 2: Client and Server (with a Real-Life Example)

- **Client:** The device or software application (like your smartphone or web browser) that requests a service or information.
- **Server:** The powerful computer that listens for those requests, processes them, and sends back the correct information.

**Real-Life Example (Ordering Food):** > Think of it like a restaurant. You are the **Client** placing an order for a burger, and the kitchen is the **Server** that receives your order, prepares the food, and sends it back to your table.

## Task 3: Comparing Desktop, Web, and Mobile Apps

### 1. Desktop Apps

- **Difference 1:** Must be downloaded and installed directly onto a computer's local operating system (like Windows or macOS).
- **Difference 2:** Usually offer high performance and can often run offline, but updates must be manually installed by the user.

### 2. Web Apps

- **Difference 1:** Accessed entirely through a web browser (like Chrome or Safari) without needing any local installation.
- **Difference 2:** They require an active internet connection to work, but they are always up-to-date because the changes happen instantly on the server.

### 3. Mobile Apps

- **Difference 1:** Specifically designed and optimized to run on smartphones and tablets (like iOS or Android devices).
- **Difference 2:** They can easily access built-in device hardware features like the camera, GPS, and push notifications.

## Task 4: 3 Databases and Their Types

| Database Name | Type (SQL / NoSQL)                                                                       |
|---------------|------------------------------------------------------------------------------------------|
| MySQL         | <b>SQL</b> (Relational database that uses tables, rows, and structured queries)          |
| MongoDB       | <b>NoSQL</b> (Non-relational database that stores data in flexible, JSON-like documents) |
| PostgreSQL    | <b>SQL</b> (Advanced relational database known for handling complex data sets)           |